

Unified Power Format (UPF) for Low Power SoC Design

Introduction:

Unified Power Format (UPF) is an IEEE standard (IEEE 1801) developed to help digital designers describe power intent independently from functional design in hardware (RTL). This separation enables efficient low-power design without cluttering RTL with power management details. UPF allows you to:

- Define power domains and associated power/ground nets
- Specify power states, switches, retention, and isolation strategies
- Support advanced low-power techniques such as power gating, multi-voltage domains, state retention, and clock gating
- Ensure consistent low-power behaviour throughout RTL, synthesis, and simulation phases

Objectives:

Understanding and applying UPF is essential for today's SoC and ASIC designers working on complex, power-sensitive digital systems.

Block Diagram and SoC Design Considerations:

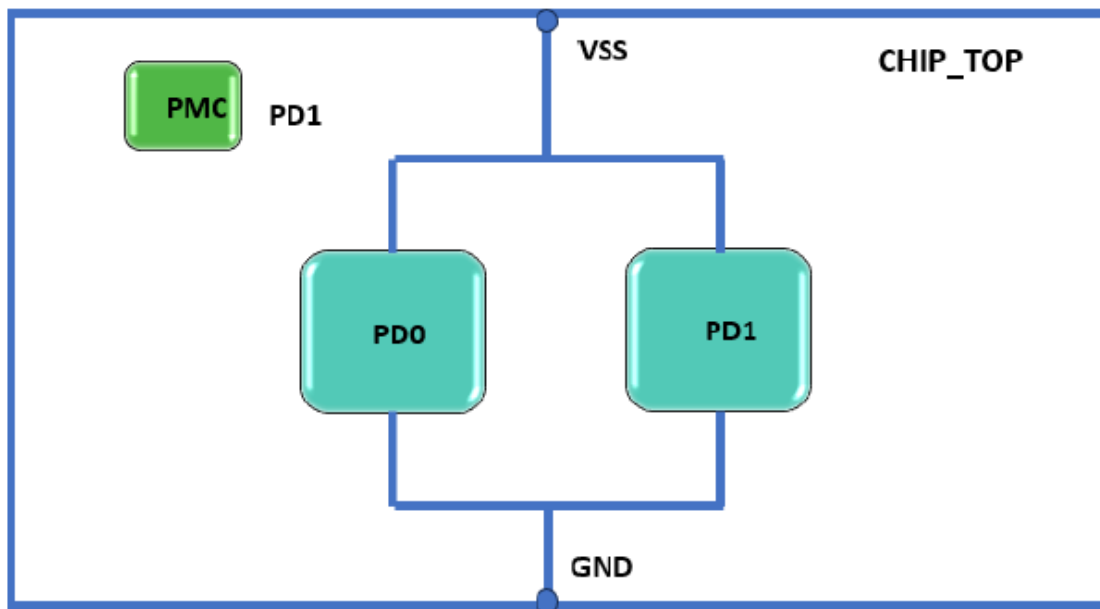


Fig: Block Diagram for power distribution

Power Domains:

A logical region of the chip that can be powered on/off independently. It is associated with:

- One or more supply nets (primary, secondary, etc.)
- A list of instances or modules it encompasses
- Power management strategies (such as isolation, retention)
- PD0 – a power domain
- PD1 – another power domain (used by PMC and another block)

PMC (Power Management Controller)

Block responsible for power sequencing, isolation, retention, and firmware interface. It is a central control block in a SoC that manages the power states of various power domains

Features

- Controls Power Domains:
- Always-On Block (in PD1)
- Manages Isolation and Retention
- Interfacing with Firmware
- PMC is in PD1

Power nets:

Logical supply lines which is used to deliver Power, Ground, and retention voltages to different power domains in a digital design

- VSS – power supply (usually logic high / main supply)
- GND – ground

CHIP_TOP:

Always-On Block: Section that must be powered even when other domains are off

- the top-level instance

Architecture of UPF and Design considerations:

An SoC (System on Chip) integrates multiple IP blocks such as CPUs, memory, I/O, and controllers often requiring:

- Different operating voltages (power domains)
- Independent clocks and resets (clock and reset domains)
- Careful interface definition for compatibility

In SoC-level design, you must manage:

- IP integration: multiple speeds, clocks, resets, and isolated functionalities
- Clock and power distribution
- Low-power techniques (clock gating, power gating, multi-Vdd)

Importance of Multiple Clock and Power Domains

- Some blocks require external clocks (e.g., USB, DDR, Ethernet), while others can run at reduced speeds to save power.
- Multiple domains help optimize each block for both performance and power efficiency.
- Example: CPUs run at high frequency, but peripherals like UARTs run much slower, reducing dynamic power consumption.

An UPF-Based Power Intent Flow overview:

1. Start with top-level domain (chip_top)
2. Define nets for all supplies and grounds
3. Create power domains for each major subsystem
4. Assign blocks/instances to proper domains
5. Specify supply sets and map to domain (allows for power gating and voltage switching)
6. PMC oversees the ON/OFF state and sequencing between these domains

Commands to create Power Domains in UPF

Step 1: Define Power Supply Nets

```
create_power_domain chip_top
create_power_net VSS -domain chip_top
create_ground_net GND -domain chip_top
```

Step 2: Create Power Domains

```
create_power_domain PD0 -elements {chip_top/PD0_inst}
create_power_domain PD1 -elements {chip_top / PD1_inst chip_top/
PMC}
```

Step 3: Create Power Supply Sets

```
create_supply_port VSS -direction in
create_supply_port GND -direction in
```

Step 4: Create Supply Sets for PD0

```
create_supply_set PD0_set -power {VSS} -ground {GND}
set_domain_supply_net PD0 -power VSS -ground GND
set_domain_supply_set PD0 PD0_set
```

Step 5: Create Supply Sets for PD1

```
create_supply_set PD1_set -power {VSS} -ground {GND}
set_domain_supply_net PD1 -power VSS -ground GND
set_domain_supply_set PD1 PD1_set
```

Applications of UPF in SoC Design

- Essential for modern microprocessors, SoCs, and embedded systems to reduce overall power.
- Used in power management units for consumer electronics, smartphones, wearables, etc.
- Critical for chips requiring standby (retention), dynamic voltage scaling, and aggressive leakage reduction.

Conclusion

Through this lab, you learned the basics of Unified Power Format (UPF) and its critical role in low-power SoC design. You now know how to formally define power domains, associate supply nets, and manage power intent separately from RTL functionality. This foundation will help you implement, simulate, and verify advanced low-power techniques in complex digital designs ensuring efficient and reliable chip operation across the full lifecycle of a semiconductor product.